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REVIEW

Bayesian Network Meta-Analysis of Postoperative Analgesic Techniques in Thoracoscopic Lung Resection Patients

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ABSTRACT

Introduction: Postoperative analgesia in thoracoscopic lung resection is crucial, with several nerve block techniques—including thoracic epidural anesthesia (TEA), paravertebral block (PVB), erector spinae plane block (ESPB), intercostal nerve block (INB), and serratus anterior plane block (SAPB)—commonly employed. However, there remains ongoing debate regarding the optimal technique.

Methods: To evaluate and compare the effectiveness of these analgesia methods, a systematic review was conducted across multiple databases, including PubMed, Embase, Web of Science, and the Cochrane Library, identifying relevant randomized clinical trials (RCTs). A Bayesian network meta-analysis was performed to assess postoperative pain management, with subgroup analyses and meta-regression conducted to examine key factors influencing outcomes, such as the risk of bias, continuous catheter analgesia, and patient-controlled analgesia (PCA).

Results: The results revealed that for 12-h resting visual analog scale (VAS) scores, the surface under the cumulative ranking curve (SUCRA) ranking was TEA > PVB > ESPB > control > INB > SAPB, whereas at 24 h, it shifted to PVB > TEA > ESPB > INB > control > SAPB. For 12-h coughing VAS scores, TEA ranked highest, followed by PVB, SAPB, ESPB, and control. At 24 h, PVB ranked highest, followed by TEA, ESPB, SAPB, INB, and control. The inconsistency test showed good consistency, with minimal publication bias, and meta-regression revealed that neither study quality nor local anesthetic infiltration at the incision site significantly impacted the outcomes. Excluding studies without PCA did not change the SUCRA rankings. PVB consistently ranked highest for 24-h resting and coughing VAS scores. Clustered ranking plots indicated that

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PVB and ESPB were the most suitable techniques for postoperative analgesia.
Conclusion: PVB and ESPB emerged as the most suitable analgesic techniques for thoracoscopic lung resection. While PVB showed superior analgesic efficacy, ESPB offered fewer side effects, providing a safety advantage. TEA was considered less suitable due to its excessive side effects.

Keywords: Thoracoscopic lung resection; TEA; PVB; ESPB; INB; SAPB

Key Summary Points
Why carry out this study?
The postoperative analgesia of thoracoscopic lung resection is crucial, However, there is still ongoing debate regarding the optimal technique.
Previous network meta-analyses have demonstrated deficiencies in clinical relevance and statistical rigor, underscoring the necessity for further refinement and improvement.
What was learned from the study?
Paravertebral block (PVB) provides superior analgesic efficacy compared to erector spinae plane block (ESPB), and ESPB is associated with fewer side effects, whereas thoracic epidural anesthesia (TEA) is less favorable due to its significant side effects.
In clinical practice, PVB is recommended as the first-line choice for postoperative analgesia in patients requiring strong pain relief, and the procedure should be performed by experienced anesthesiologists.
For patients with coagulation disorders or those at high risk for complications from nerve blocks, or when the anesthesiologist has limited experience with nerve block techniques, ESPB may be a safer and more appropriate alternative.

INTRODUCTION

With advancements in technology, surgery is progressively becoming more minimally invasive and pain-free. Thoracoscopic lung resection, including lobectomy, segmentectomy, and pleural biopsy, has increasingly gained prominence in lung surgery. Although thoracoscopic procedures are minimally invasive, severe postoperative pain remains a persistent and challenging complication, which may hinder patient recovery. The primary sources of this pain include the surgical incision, intercostal nerve irritation caused by surgical instruments, and pleural traction, all of which contribute to patient discomfort [1]. This pain adversely affects patient comfort, restricts deep breathing, and consequently elevates the risk of atelectasis and pneumonia, ultimately extending recovery duration.

Postoperative pain management relies predominantly on pharmacological interventions, including opioids and nonsteroidal anti-inflammatory drugs (NSAIDs). However, the significant limitations of pharmacological analgesia following thoracoscopic thoracic surgery—such as nausea, vomiting, sedation, addiction risks, gastrointestinal side effects, and potential renal impairment—have led clinicians to increasingly seek safer and more effective non-pharmacological alternatives. Recent advancements in ultrasound-guided nerve block techniques have significantly enhanced their prominence in clinical research, primarily due to their efficacy in managing postoperative pain. Historically, thoracic epidural anesthesia (TEA) and paravertebral block (PVB) have been the principal methods for postoperative analgesia following thoracic surgery, both widely recognized and utilized in clinical settings. Additionally, serratus anterior plane block (SAPB) and intercostal nerve block (INB) have proven effective in managing pain after thoracic procedures [2, 3]. Notably, in 2018, ESPB was first documented in the literature for successfully alleviating postoperative pain in three patients undergoing thoracoscopic surgery [4]. Following this, the *Journal of Clinical Anesthesia* in 2019 published further case reports detailing the application of ESPB in post-thoracoscopic surgical analgesia [5]. A

previous network meta-analysis assessed postoperative pain management methods in thoracic surgery. However, the study focused primarily on pain after lung resection and included data from both open thoracotomy and thoracoscopic lung resection surgeries, which is methodologically inappropriate [6]. It is well established that postoperative pain intensity significantly differs between patients undergoing open surgery and those undergoing thoracoscopic surgery [7]. Additionally, another study compared postoperative pain management across various thoracoscopic procedures, including lung resection, esophagectomy, mediastinal tissue resection, and even cardiac surgery. The thoracoscopic approaches used for these procedures varied considerably—some involved intercostal access, others used a subxiphoid approach, and some employed a parasternal approach—resulting in substantial differences in both the location and intensity of postoperative pain [8].

At present, prevalent ultrasound-guided techniques for postoperative analgesia in thoracic surgery encompass TEA, PVB, SAPB, INB, and ESPB [9]. While these techniques have demonstrated effectiveness in studies, their relative advantages remain uncertain. To explore the optimal non-pharmacological pain management strategies following thoracoscopic lung resection, this study is based on 28 RCTs and systematically evaluates the effectiveness and safety of various regional nerve block techniques, as well as commonly used PCA, in postoperative pain management following thoracoscopic lung resection. Additionally, the study analyzes the impact of various factors influencing pain scores on the outcomes, aiming to provide clear recommendations and identify the optimal approach for managing postoperative pain in thoracoscopic lung resection.

METHODS

Literature Search Strategy

This study follows the PRISMA (Preferred Reporting Items for Systematic reviews and Meta-Analyses) guidelines and AMSTAR (A Measurement Tool to Assess Systematic Reviews) standards to ensure methodological rigor. This article does not contain any new studies with human participants or animals performed by any of the authors. A comprehensive search was performed across PubMed, Embase, Web of Science, and the Cochrane Library, covering records from inception through September 2024. Relevant keywords and subject headings included the following: thoracic paravertebral block; continuous thoracic paravertebral block; erector spinae plane block; continuous erector spinae block; intercostal nerve block; serratus anterior plane block; epidural analgesia; nerve block; Block, Nerve; Blocks, Nerve; Nerve Blocks; Chemical Neurolysis; Chemical Neurolyses; Neurolyses, Chemical; Neurolysis, Chemical; Chemodenervation; Chemodenervations; Nerve Blockade; Blockade, Nerve; Blockades, Nerve; Nerve Blockades; Thoracoscopic Lobectomy; Thoracoscopic surgery lobectomy; Thoracic Surgery, Video-Assisted; Surgeries, Video-Assisted Thoracic; Surgery, Video-Assisted Thoracic; Thoracic Surgeries, Video-Assisted; Thoracic Surgery, Video Assisted; Video-Assisted Thoracic Surgeries; Surgery, Thoracic, Video-Assisted; VATS; VATSs; Video-Assisted Thoracic Surgery.

Inclusion and Exclusion Criteria

The inclusion criteria were as follows: (1) patients undergoing thoracoscopic lung resection surgery, including lobectomy, pneumonectomy, segmentectomy, or other procedures; (2) studies evaluating at least one of the following interventions: SAPB, ESPB, PVB, TEA, or INB; (3) articles reporting one or more of the following outcomes: pain scores at rest or during movement, adverse events (e.g., nausea, vomiting, pruritus, dizziness), cumulative 24-h opioid consumption, postoperative complications (e.g., pneumonia, pneumothorax, atelectasis), or length of hospital stay; and (4) RCTs. Articles were excluded if they were evaluations, observational studies, letters, case reports, or animal research.

Study Selection and Data Extraction

Two researchers independently reviewed and assessed the full texts to identify studies that met the inclusion criteria. Data extraction was performed by one researcher and independently verified by another. Discrepancies were resolved through re-examination of the manuscripts, with a third researcher consulted for further evaluation if necessary. During data collection, we recorded the following primary outcomes: postoperative VAS scores at various time points (including resting and coughing VAS), length of hospital stays, and postoperative complications (e.g., nausea, vomiting, atelectasis, dizziness, headache, pulmonary infections, and chest tube removal time). We also gathered data on the use and dosage of rescue analgesics postoperatively and the intraoperative administration of analgesic medications. Continuous variables were reported as means and standard deviations, while categorical variables were presented as frequencies. For RCTs reporting data as medians and interquartile ranges, means and standard deviations were estimated using sample sizes [10]. For studies lacking specific research data, in addition to contacting the authors for information, graphical data were extracted by creating a simulated coordinate system using the GetData Graph Digitizer software (version 2.26.12). Additionally, postoperative analgesic use was standardized by converting various opioid drugs to intravenous morphine equivalents according to published guidelines. Conversion ratios for equivalence were as follows: morphine: tramadol: pethidine: oxycodone: butorphanol: pentazocine = 1 mg: 10 mg: 7.5 mg: 0.5 mg: 0.2 mg: 6 mg [11].

Quality Assessment

The risk of bias in the included studies was assessed using Review Manager software (RevMan 5.3). The quality of each study was evaluated based on several criteria, including randomization methods, allocation concealment, blinding of participants and personnel, blinding of outcome assessors, selective reporting, incomplete outcome data, and other potential sources of bias.

Statistical Analyses

This study utilized R software (version 4.0.2) with the gemtc and netmeta packages to perform the analysis. Network evidence diagrams were created for each outcome measure, and a Bayesian node-splitting method was applied to evaluate model inconsistency using Markov chain Monte Carlo (MCMC) techniques. When $P > 0.05$, a consistency model and random-effects model were used for the network meta-analysis (NMA), with four chains, 50,000 iterations, and a burn-in of 20,000 iterations per chain. Model convergence was assessed using the Brooks–Gelman–Rubin method, with potential scale reduction factor (PSRF) values close to 1 considered indicative of satisfactory convergence. For binary outcomes, effect estimates were expressed as odds ratios (OR), while continuous outcomes were reported as mean differences (MD).

In the Bayesian network meta-analysis, we used prior distributions for both treatment effects and heterogeneity. For treatment effects (e.g., differences in pain scores), we applied uninformative normal priors with a mean of zero and a standard deviation of 1. This ensures that the results are primarily driven by the data, not by prior assumptions. For the heterogeneity parameter (τ), we used a gamma distribution based on variability observed in similar studies. Sensitivity analyses showed that the results were consistent across different prior choices.

Heterogeneity was evaluated using the I^2 statistic, with values exceeding 50% indicating substantial heterogeneity among studies. To address high heterogeneity, this study performed subgroup analysis, meta-regression, and sensitivity analysis to identify potential sources. Publication bias was evaluated using funnel plots and Egger's test.

RESULTS

Baseline Characteristics and Risk of Bias

The initial search yielded 350 studies, of which 28 randomized controlled trials were included. These trials involved a total of 2185 patients and evaluated six different analgesic intervention strategies (TEA, PVB, ESPB, SAPB, INB, control). The search process is summarized in Supplementary Material–Supplement 1.

Among the included 28 trials [12–39], five were three-arm studies, while the majority were two-arm trials. Among these, eight clinical trials involved SAPB, 13 included PVB, and eight examined ESPB. Additionally, four clinical trials investigated INB, and five included TEA. A total of 20 clinical trials included the control group. Baseline characteristics are presented in Supplementary Material–Supplement 2. Among the 28 studies included, seven were classified as high risk, and 18 as moderate risk. For further details, refer to Supplementary Material–Supplement 3.

Primary Outcomes

Resting VAS Scores at Postoperative 12, 24, and 48 Hours

Among the 28 clinical trials, 14 studies, including 1108 patients, reported postoperative resting VAS scores at 12 h. The SUCRA ranking was TEA > PVB > ESPB > control > INB > SAPB. However, pairwise comparisons showed no significant differences among these six postoperative analgesic techniques. For postoperative resting VAS scores at 24 h, 25 trials involving 1917 patients were included. The SUCRA ranking was PVB > TEA > ESPB > INB > control > SAPB, with the league table indicating significant statistical differences between SAPB and PVB (–1.1 [–2.17, –0.05]) and between PVB and the control group (1.01 [0.38, 1.65]). Additionally, 19 studies, comprising 1542 patients, reported postoperative resting VAS scores at 48 h. The SUCRA ranking was TEA > PVB > ESPB > SAPB > INB > control, with pairwise comparisons revealing significant statistical differences between the control group and TEA (–0.79 [–1.38, –0.17]) and between PVB and the control group (0.61 [0.21, 1.01]). Finally, the PSRF values for the Bayesian models were all approximately 1, indicating excellent model convergence (Refer to Tables 1, 2; Figs. 1, 2).

Coughing VAS Scores at Postoperative 12, 24, and 48 Hours

Among the 28 included clinical trials, 9, 17, and 14 studies assessed postoperative coughing VAS scores at 12, 24, and 48 h, respectively, with sample sizes of 855, 1503, and 1265 patients. The SUCRA ranking for 12-h coughing VAS scores was TEA > PVB > SAPB > ESPB > control. Pairwise comparisons revealed significant statistical differences between SAPB and control (1.16 [0.15, 2.16]), PVB and control (1.23 [0.54, 1.93]), control and ESPB (–0.81 [–1.57, –0.05]), and control and TEA (–2.73 [–4.46, –1.01]). For 24-h postoperative coughing VAS scores, the SUCRA ranking was PVB > TEA > ESPB > SAPB > INB > control, with pairwise comparisons showing a significant difference between PVB and control (0.95 [0.32, 1.59]). At 48 h postoperatively, the SUCRA ranking was PVB > TEA > ESPB > SAPB > control > INB. However, pairwise comparisons among these six analgesic techniques showed no statistically significant differences. Additionally, the Bayesian network meta-analysis for these outcomes demonstrated PSRF values of 1, confirming excellent model convergence and fit (refer to Tables 1, 2; Figs. 1, 2).

Comparison of Adverse Events and Length of Hospital Stay

A total of 21 studies, comprising 1669 patients, reported data on postoperative nausea and vomiting (PONV). The SUCRA rankings for this outcome were as follows: ESPB > PVB > SAPB > INB > TEA > control. Regarding postoperative pulmonary complications, nine studies involving 816 patients provided data, with the ranking being ESPB > PVB > TEA > control > SAPB. Furthermore, 12 studies, including 1004 patients, analyzed the length of hospital stay, with the SUCRA rankings indicating

Table 1 SUCRA ranking for the primary outcomes

Postoperative 12-h resting VAS			Postoperative 24-h resting VAS			Postoperative 48-h resting VAS		
14 trials, 1108 patients, PSRF = 1			25 trials, 1917 patients, PSRF = 1.01			19 trials, 1542 patients, PSRF = 1		
Rank	Treatment	SUCRA	Rank	Treatment	SUCRA	Rank	Treatment	SUCRA
1	TEA	0.828	1	PVB	0.839	1	TEA	0.884
2	PVB	0.745	2	TEA	0.810	2	PVB	0.757
3	ESPB	0.587	3	ESPB	0.628	3	ESPB	0.519
4	Control	0.369	4	INB	0.289	4	SAPB	0.452
5	INB	0.283	5	Control	0.232	5	INB	0.245
6	SAPB	0.188	6	SAPB	0.201	6	Control	0.144
Postoperative 12-h coughing VAS			**Postoperative 24-h coughing VAS**			**Postoperative 48-h coughing VAS**		
9 trials, 855 patients, PSRF = 1			17 trials, 1503 patients, PSRF = 1			14 trials, 1265 patients, PSRF = 1		
Rank	Treatment	SUCRA	Rank	Treatment	SUCRA	Rank	Treatment	SUCRA
1	TEA	0.974	1	PVB	0.865	1	PVB	0.789
2	PVB	0.613	2	TEA	0.709	2	TEA	0.697
3	SAPB	0.561	3	ESPB	0.581	3	ESPB	0.608
4	ESPB	0.342	4	SAPB	0.324	4	SAPB	0.513
5	Control	0.010	5	INB	0.303	5	Control	0.254
			6	Control	0.218	6	INB	0.138
Postoperative nausea and vomiting			**Pulmonary complications**			**Length of hospital stay**		
21 trials, 1669 patients, PSRF = 1			9 trials, 816 patients, PSRF = 1			12 trials, 1004 patients, PSRF = 1		
Rank	Treatment	SUCRA	Rank	Treatment	SUCRA	Rank	Treatment	SUCRA
1	ESPB	0.846	1	ESPB	0.692	1	INB	0.781
2	PVB	0.719	2	PVB	0.655	2	ESPB	0.678
3	SAPB	0.456	3	TEA	0.501	3	PVB	0.507
4	INB	0.451	4	Control	0.368	4	TEA	0.489
5	TEA	0.379	5	SAPB	0.265	5	SAPB	0.415
6	Control	0.149				6	Control	0.129

INB > ESPB > PVB > TEA > SAPB > control. The league table revealed significant statistical differences in PONV between PVB and the control group (odds ratio [OR]: 2.77; 95% confidence interval [CI]: 1.2–6.4) and between the control and ESPB (OR: 0.27; 95% CI: 0.1–0.76) (Refer to Tables 1, 2; Figs. 1, 2).

Secondary Outcomes

Additionally, postoperative rescue analgesic consumption was reported in six of the 28 trials, encompassing 356 patients. The analysis yielded the ranking: TEA (0.814) > PVB (0.687) > ESPB (0.598) > SAPB (0.296) > control

Table 2 Pairwise comparison of primary outcome measures (league table)

Postoperative 12-h resting VAS						Postoperative 12-h coughing VAS					
SAPB	**−1.66 (−3.87, 0.54)**	**−0.72 (−2.5, 1.08)**	**−1.21 (−3.26, 0.83)**	**0.00 (−3.19, 3.19)**	**−2.17 (−5.16, 0.82)**	**SAPB**	**−0.07 (−1.25, 1.12)**	**1.16 (0.15, 2.16)**	**0.35 (−0.76, 1.46)**	**−1.56 (−3.55, 0.39)**	
	PVB	0.95 (−0.42, 2.31)	0.45 (−1.17, 2.05)	1.65 (−2.18, 5.55)	−0.52 (−2.92, 1.87)		**PVB**	**1.23 (0.54, 1.93)**	0.42 (−0.48, 1.25)	−1.5 (−3.08, 0.08)	
		Control	−0.50 (−1.79, 0.79)	0.71 (−2.92, 4.38)	−1.47 (−3.88, 0.95)			**Control**	**−0.81 (−1.57, −0.05)**	**−2.73 (−4.46, −1.01)**	
			ESPB	1.22 (−2.56, 5.01)	−0.96 (−3.61, 1.68)				**ESPB**	**−1.91 (−3.70, −0.15)**	
				INB	−2.17 (−6.55, 2.19)					**TEA**	
					TEA						
Postoperative 24-h resting VAS						**Postoperative 24-h coughing VAS**					
SAPB	**−1.1 (−2.17,−0.05)**	−0.09 (−0.98, 0.78)	−0.72 (−1.83, 0.38)	−0.13 (−1.49, 1.22)	−1.1 (−2.42, 0.23)	**SAPB**	−0.87 (−1.98, 0.17)	0.08 (−0.75, 0.95)	−0.41 (−1.49, 0.68)	0.04 (−1.78, 1.79)	−0.67 (−2.05, 0.73)
	PVB	**1.01 (0.38, 1.65)**	0.38 (−0.56, 1.34)	0.97 (−0.40, 2.36)	0.00 (−1.04, 1.07)		**PVB**	**0.95 (0.32, 1.59)**	0.46 (−0.43, 1.34)	0.90 (−0.79, 2.54)	0.19 (−0.86, 1.26)
		Control	−0.63 (−1.48, 0.23)	−0.04 (−1.34, 1.28)	−1.01 (−2.05, 0.05)			**Control**	−0.49 (−1.33, 0.33)	−0.05 (−1.72, 1.56)	−0.75 (−1.87, 0.36)

Table 2 continued

Postoperative 12-h resting VAS						**Postoperative 12-h coughing VAS**					
SAPB	**−1.66 (−3.87, 0.54)**	**−0.72 (−2.5, 1.08)**	**−1.21 (−3.26, 0.83)**	**0.00 (−3.19, 3.19)**	**−2.17 (−5.16, 0.82)**	**SAPB**	**−0.07 (−1.25, 1.12)**	**1.16 (0.15, 2.16)**	**0.35 (−0.76, 1.46)**	**−1.56 (−3.55, 0.39)**	
			ESPB	0.59 (−0.66, 1.85)	−0.38 (−1.64, 0.89)				**ESPB**	0.45 (−1.00, 1.83)	−0.26 (−1.57, 1.06)
				INB	−0.97 (−2.45, 0.51)					**INB**	−0.71 (−2.60, 1.26)
					TEA						**TEA**
Postoperative 48-h resting VAS						**Postoperative 48-h coughing VAS**					
SAPB	−0.32 (−1.07, 0.45)	0.29 (−0.34, 0.95)	−0.07 (−0.91, 0.76)	0.26 (−0.85, 1.39)	−0.50 (−1.36, 0.41)	**SAPB**	−0.46 (−1.91, 1.01)	0.38 (−0.75, 1.57)	−0.17 (−1.8, 1.51)	0.95 (−1.26, 3.3)	−0.37 (−2.09, 1.43)
	PVB	**0.61 (0.21, 1.01)**	0.25 (−0.32, 0.79)	0.58 (−0.35, 1.52)	−0.18 (−0.75, 0.43)		**PVB**	0.84 (−0.02, 1.73)	0.30 (−0.86, 1.47)	1.42 (−0.49, 3.42)	0.09 (−1.11, 1.34)
		Control	−0.36 (−0.91, 0.15)	−0.04 (−0.95, 0.89)	**−0.79 (−1.38, −0.17)**			**Control**	−0.54 (−1.72, 0.63)	0.57 (−1.33, 2.57)	−0.75 (−2.05, 0.59)
			ESPB	0.33 (−0.48, 1.17)	−0.43 (−1.13, 0.33)				**ESPB**	1.12 (−0.41, 2.72)	−0.20 (−1.80, 1.43)
				INB	−0.76 (−1.71, 0.22)					**INB**	−1.32 (−3.60, 0.89)
					TEA						**TEA**

Table 2 continued

Postoperative 12-h resting VAS						Postoperative 12-h coughing VAS				
SAPB	**−1.66 (−3.87, 0.54)**	**−0.72 (−2.5, 1.08)**	**−1.21 (−3.26, 0.83)**	**0.00 (−3.19, 3.19)**	**−2.17 (−5.16, 0.82)**	**SAPB**	**−0.07 (−1.25, 1.12)**	**1.16 (0.15, 2.16)**	**0.35 (−0.76, 1.46)**	**−1.56 (−3.55, 0.39)**
Postoperative nausea and vomiting						**Pulmonary complications**				
SAPB	0.62 (0.15, 2.62)	1.71 (0.52, 5.71)	0.47 (0.11, 2.08)	1.01 (0.12, 9.36)	1.15 (0.26, 7.11)	**SAPB**	0.54 (0.12, 2.49)	0.78 (0.23, 2.64)	0.46 (0.08, 2.20)	0.66 (0.08, 5.77)
	PVB	**2.77 (1.20, 6.40)**	0.76 (0.24, 2.45)	1.63 (0.24, 11.87)	1.85 (0.53, 7.09)		**PVB**	1.43 (0.60, 3.49)	0.85 (0.11, 5.34)	1.21 (0.26, 5.77)
		Control	**0.27 (0.10, 0.76)**	0.59 (0.09, 4.03)	0.67 (0.18, 2.69)			**Control**	0.59 (0.09, 2.93)	0.84 (0.15, 5.05)
			ESPB	2.14 (0.38, 12.4)	2.44 (0.53, 11.75)				**ESPB**	1.47 (0.13, 18.30)
				INB	1.14 (0.15, 8.82)					**TEA**
					TEA					
Length of hospital stay										
SAPB	−0.23 (−2.27, 1.95)	0.55 (−1.18, 2.33)	−0.68 (−3.12, 1.80)	−1.2 (−4.31, 2.00)	−0.2 (−2.49, 2.14)					
	PVB	0.78 (−0.44, 1.89)	−0.46 (−2.26, 1.27)	−0.97 (−3.69, 1.70)	0.03 (−1.64, 1.64)					
		Control	−1.23 (−2.95, 0.49)	−1.75 (−4.39, 0.91)	−0.75 (−2.25, 0.76)					

Table 2 continued

	Postoperative 12-h resting VAS						Postoperative 12-h coughing VAS			
	SAPB	**−0.72 (−2.5, 1.08)**	**−1.21 (−3.26, 0.83)**	**0.00 (−3.19, 3.19)**	**−2.17 (−5.16, 0.82)**	**−1.66 (−3.87, 0.54)**	**SAPB**	**−0.07 (−1.25, 1.12)**	**1.16 (0.15, 2.16)**	**0.35 (−0.76, 1.46)**	**−1.56 (−3.55, 0.39)**
			ESPB	−0.52 (−2.91, 1.92)	0.48 (−1.59, 2.55)						
				INB	1.00 (−1.67, 3.65)						
					TEA						

(0.105) (Supplementary Material–Supplement 4).

Inconsistency, Heterogeneity, and Bias

The results of inconsistency and heterogeneity tests for the primary outcomes are summarized in Table 3, with detailed findings provided in the Supplementary Material–Supplement 5. Most loop inconsistency tests had $P>0.05$, indicating overall consistency, though a few loops demonstrated inconsistency. Conversely, the heterogeneity tests revealed relatively high levels of heterogeneity. Publication bias was identified only in the 12-h coughing VAS and nausea and vomiting outcomes, while no bias was detected in the other outcomes (Supplementary Material–Supplement 6). The model fitting performance and convergence of all primary outcomes were satisfactory (refer to Supplementary Material–Supplement 7).

Subgroup Analyses and Sensitivity Analyses

Comparison Following the Exclusion of Clinical Trials Without PCA Implementation

Of the 28 included clinical trials, eight trials with control groups that did not utilize PCA were excluded. This step ensured that all control groups consistently employed PCA for postoperative analgesia. The remaining 20 trials were analyzed to compare postoperative analgesic methods, including SAPB, PVB, TEA, ESPB, PCA, and INB, by evaluating resting and coughing VAS scores at 12, 24, and 48 h postoperatively. Additionally, the analysis considered outcomes such as postoperative nausea and vomiting (PONV), length of hospital stays, and other adverse events.

The SUCRA rankings for the postoperative 12-h resting and coughing VAS remained unchanged. For the 24-h resting VAS and the 24-h and 48-h coughing VAS, PVB consistently achieved the highest SUCRA ranking, consistent with the results prior to exclusions. However, for the 48-h resting VAS, the rankings of PVB and TEA were reversed. In the SUCRA ranking

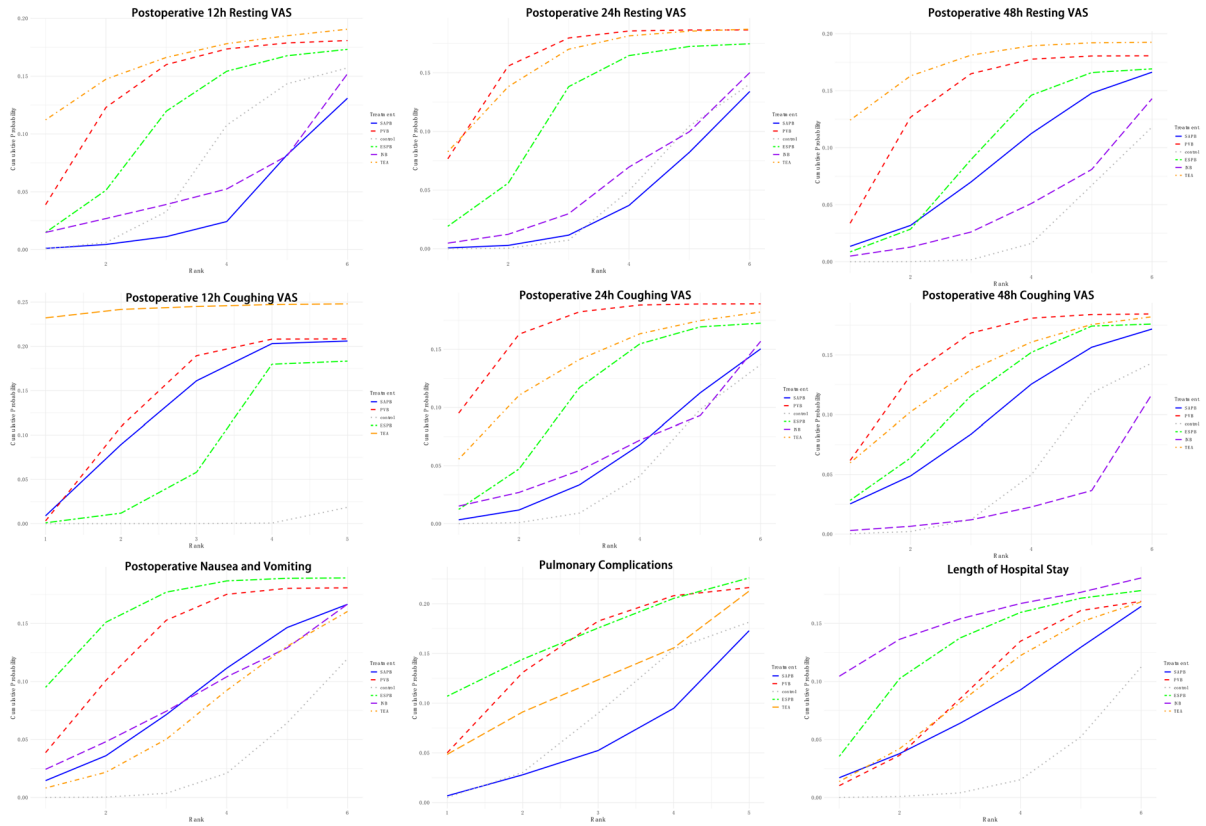

Fig. 1 SUCRA ranking plot for primary outcome measures

table provided in Supplementary Material–Supplement 8, orange-highlighted sections denote interventions whose rankings changed compared to previous results.

The league table revealed significant statistical differences between PVB and PCA (odds ratio [OR]: 0.46; 95% confidence interval [CI] 0.14–0.77). For PONV, ESPB maintained the highest ranking, with significant statistical differences observed between ESPB and PCA (OR: 0.3; 95% CI 0.13–0.77) and between ESPB and TEA (OR: 8.67; 95% CI 1.53–41.18). Regarding length of hospital stay, PCA had a substantial impact, leading to notable shifts in rankings (see Supplementary Material–Supplement 8).

Comparison Following the Exclusion of High-Risk Clinical Trials

Among the 28 included studies, seven were identified as having high-risk bias factors. After excluding these studies, a network meta-analysis was conducted to evaluate the primary outcomes. The analysis showed that while the SUCRA rankings for the postoperative 12-h coughing VAS remained unchanged, minor variations were observed in other rankings. For the postoperative 12-h resting VAS and 24-h resting VAS, the top three techniques (TEA, PVB, and ESPB) retained their positions. Additionally, the league table demonstrated significant statistical differences between TEA, PVB, ESPB, and the control group for the 12-h resting VAS. Similarly, for the 24-h resting VAS, both PVB and ESPB exhibited significant statistical differences compared to the control group (see Supplementary Material–Supplement 9).

For the 24-h coughing VAS, PVB maintained the highest ranking, while the rankings for ESPB and TEA shifted. The league table highlighted a significant statistical difference

Fig. 2 Network evidence diagram for primary outcome measures

between PVB and the control group. Regarding the 48-h dynamic VAS, SUCRA rankings displayed some changes, with the league table indicating that only the 48-h resting VAS showed a significant statistical difference between PVB and the control group (see Supplementary Material–Supplement 9). In terms of postoperative nausea and vomiting, ESPB retained the highest SUCRA ranking and demonstrated a significant statistical difference compared to the control group. Rankings for length of hospital stay remained largely unchanged (see Supplementary Material–Supplement 9).

Comparison Following the Exclusion of Postoperative Continuous Catheter-Based Nerve Blocks

The use of catheters in nerve blocks is a key factor affecting VAS scores. A comprehensive review of 28 clinical trials identified five studies that utilized continuous catheter insertion techniques in specific groups. To ensure consistency, only trials employing single-injection nerve blocks were included, leading to the exclusion of these five studies. As a result, 23 trials were retained for analysis. These trials were then analyzed to compare six postoperative analgesic techniques based on primary outcome measures.

Table 3 Inconsistency and heterogeneity tests for the primary outcomes

Postoperative 24-h resting VAS				
	Node-splitting analysis of inconsistency			**Heterogeneity**
Comparison	**Direct**	**Indirect**	***p***	I^2(%)
SAPB vs. control	−0.12 (−1.21, 0.88)	0.10 (−2.70, 2.91)	0.87	95.32
SAPB vs. ESPB	−0.50 (−2.70, 1.73)	−0.80 (−2.22, 0.53)	0.81	0.00
SAPB vs. INB	−0.01 (−2.20, 2.10)	−0.23 (−2.13, 1.57)	0.86	94.52
PVB vs. control	1.03 (0.29, 1.82)	0.87 (−0.81, 2.63)	0.85	90.40
PVB vs. ESPB	0.07 (−1.53, 1.67)	0.53 (−0.76, 1.82)	0.64	16.12
PVB vs. TEA	0.04 (−1.51, 1.65)	−0.01 (−1.60, 1.62)	0.96	91.63
Control vs. ESPB	−0.51 (−1.61, 0.68)	−0.90 (−2.71, 0.88)	0.69	68.43
Control vs. TEA	−1.13 (−2.73, 0.45)	−0.90 (−2.41, 0.60)	0.82	87.15
ESPB vs. INB	0.62 (−1.12, 2.38)	0.55 (−1.42, 2.68)	0.96	0.00
INB vs. TEA	−0.67 (−3.31, 1.91)	−1.14 (−3.32, 0.75)	0.79	0.00
Postoperative nausea and vomiting				
	Node-splitting analysis of inconsistency			**Heterogeneity**
Comparison	**Direct**	**Indirect**	***p***	I^2(%)
SAPB vs. ESPB	0.34 (−2.12, 2.82)	−1.31 (−3.17, 0.59)	0.28	45.52
PVB vs. control	0.79 (−0.08, 1.71)	2.03 (−0.07, 3.92)	0.27	44.23
PVB vs. ESPB	−0.62 (−2.61, 1.34)	−0.08 (−1.60, 1.62)	0.66	64.21
PVB vs. TEA	1.42 (0.03, 2.91)	−0.88 (−2.61, 1.18)	0.06	0.00
Control vs. ESPB	−1.10 (−2.35, 0.20)	−2.2 (−4.92, 0.35)	0.44	77.13
Control vs. TEA	−2.24 (−4.28, −0.36)	0.65 (−0.76, 2.1)	0.02	45.02
ESPB vs. INB	0.99 (−0.97, 3.12)	−0.20 (−4.52, 3.55)	0.57	92.31
INB vs. TEA	0.92 (−2.43, 4.91)	−0.29 (−3.03, 2.32)	0.57	0.00

The SUCRA rankings for postoperative 12-h resting VAS, 12-h coughing VAS, 24-h resting VAS, and 48-h coughing VAS scores remained consistent with the results prior to exclusions. For the 24-h coughing VAS scores, PVB maintained the highest ranking, with the league table indicating a significant statistical difference between PVB and the control group (0.87 [0.13, 1.61]). Similarly, for the 48-h resting VAS scores, PVB ranked first, with the league table showing a significant statistical difference compared to the control group (0.52 [0.2, 0.83]). For postoperative nausea and vomiting, ESPB retained the highest ranking, with the league table showing a significant statistical difference between ESPB and the control group (0.32 [0.12, 0.93]). In terms of length of hospital stay, catheterization appeared to have a considerable influence, leading to a notable shift in rankings (see Supplementary Material–Supplement 10).

Meta-Regression Analyses

Among the 28 included studies, some employed local anesthetic infiltration (LA) at the incision site in addition to nerve block analgesia. A meta-regression analysis was performed to evaluate the impact of LA on the outcomes, revealing no statistically significant effect (Supplementary Material–Supplement 11). Furthermore, the influence of the risk of bias (RoB) in the included studies was assessed, showing no overall effect on most outcomes (Supplementary Material–Supplement 12). However, for the 48-h resting VAS, RoB was found to negatively affect ESPB: INB (–1.91 [95% CI: –3.33, –0.48]). Similarly, for postoperative nausea and vomiting, RoB had a negative impact on PVB: TEA (–1.85 [95% CI: –3.43, –0.33]).

Clustered Ranking Plots

Given the clinical significance and the stability of the study results, this research selected postoperative 12-h and 24-h resting VAS scores, as well as 12-h and 24-h coughing VAS scores, to perform cluster ranking for nausea and vomiting outcomes (Fig. 3). The results show that PVB and ESPB are located near the upper-right corner of the cluster ranking chart. Additionally, cluster analysis was conducted on 24-h postoperative

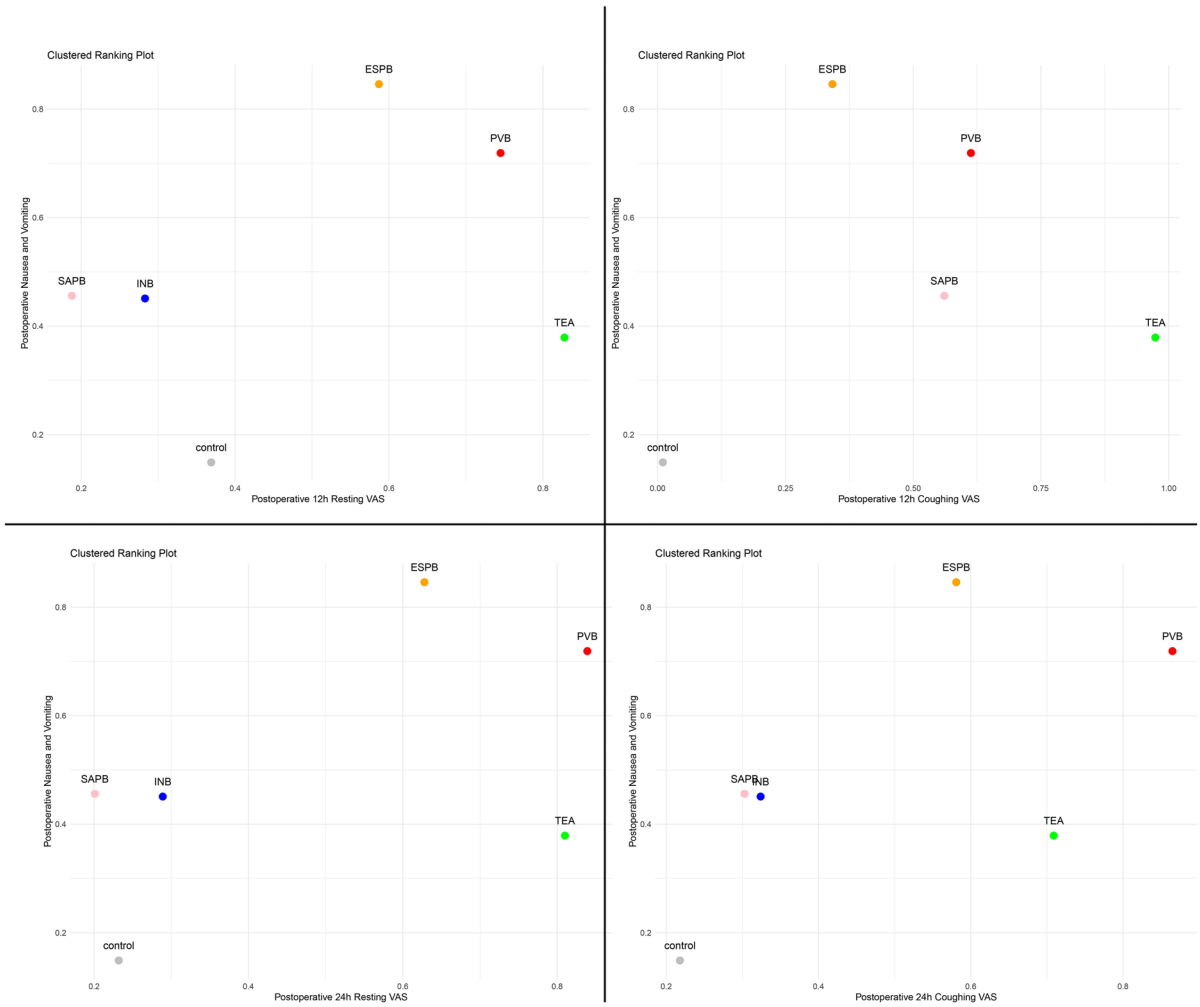

Fig. 3 Two-dimensional clusters plot of primary outcome measures and postoperative nausea and vomiting

VAS scores (resting and coughing) and other adverse effect dimensions, including pulmonary complications and length of hospital stay. This analysis aimed to identify the analgesic technique that offers the most comprehensive advantages across multiple adverse effect dimensions (Fig. 4).

DISCUSSION

This meta-analysis incorporated inconsistency testing and publication bias assessment, ensuring a high degree of reliability in its findings. While the network meta-analysis identified substantial heterogeneity in some indicators, these issues were addressed through subgroup analyses and meta-regression. These approaches systematically evaluated key factors influencing postoperative pain outcomes, including the risk of bias (RoB), the use of continuous catheter-based analgesia, and the implementation of PCA. As a result, stable indicators were identified, such as the 12-h and 24-h postoperative resting VAS, 12-h and 24-h postoperative coughing VAS, and postoperative nausea and vomiting. Clustering analysis based on these stable indicators provided robust and reliable conclusions.

The findings indicate that while TEA provides superior analgesic efficacy at 12 h

Fig. 4 Two-dimensional cluster plot combining 24-h postoperative vas scores with multiple adverse effect indicators

postoperatively, PVB surpasses TEA in analgesic effectiveness at 24 and 48 h postoperatively, for both resting and coughing VAS scores. However, TEA is associated with significantly higher rates of side effects, including nausea, vomiting, prolonged hospital stays, and pulmonary complications, making it a less favorable option for postoperative analgesia in patients undergoing thoracoscopic lung resection.

A comprehensive analysis of VAS scores at various time points and associated side effects—including nausea, vomiting, hospital length of stay, and pulmonary complications—highlighted that both PVB and ESPB are effective postoperative analgesic techniques. While PVB demonstrated superior analgesic efficacy, ESPB offered a better safety profile, with fewer cases of nausea, vomiting, pulmonary complications, and shorter hospital stays. Additionally, secondary outcomes revealed that patients receiving PVB required fewer postoperative rescue analgesics compared to those treated with ESPB.

Pain is commonly reported within the first 1–2 days post-VATS, primarily caused by surgical factors such as mechanical trauma, nerve traction or compression, and unrecognized nerve transection [40]. Preemptive analgesia for acute postoperative pain management in patients undergoing video-assisted thoracoscopic surgery (VATS) lobectomy effectively reduces opioid consumption and minimizes postoperative complications [41]. TEA has long been recognized for its effectiveness in managing postoperative pain in thoracic surgery. However, its extensive side effects, including nausea, vomiting, pruritus, urinary retention, and hypotension, have contributed to its declining use [42]. PVB, a nerve block technique involving the injection of local anesthetic into the paravertebral space, effectively blocks thoracic spinal nerves, their branches, and the sympathetic trunk. The anesthetic spreads cranially and caudally through loose connective tissue in the paravertebral space and laterally into the intercostal and epidural spaces, achieving an analgesic effect comparable to thoracic segmental epidural blockade [43, 44]. Similarly, ESPB provides analgesia by targeting the dorsal and ventral branches of the thoracic spinal nerves, desensitizing the anterior, lateral, and posterior thoracic walls [45, 46]. In contrast, SAPB involves injecting local anesthetics deep to or superficial to the serratus anterior muscle, targeting the lateral cutaneous branches of the intercostal nerves, the thoracodorsal nerve, and the long thoracic nerve, depending on the injection site [47].

The findings of this research confirm that PVB continues to provide superior analgesic efficacy compared to ESPB in patients undergoing thoracoscopic lung resection. This result is consistent with prior meta-analyses directly comparing PVB and ESPB, which have uniformly concluded that ESPB cannot fully replace PVB for postoperative analgesia [48]. The superior efficacy of PVB may be due to its effective diffusion of local anesthetics to the dorsal and ventral branches of the spinal nerves. Additionally, some studies have noted that ESPB requires a longer onset time and has inconsistent blocking efficacy, which could limit its effectiveness relative to PVB [49]. Two previous network meta-analyses concluded that ESPB could fully replace PVB, a conclusion that contradicts the findings of our study. This inconsistency may stem from methodological differences. One study combined open thoracotomy and thoracoscopic procedures into a single statistical analysis, while the other included a wide range of thoracoscopic surgeries—such as mediastinal procedures, esophagectomy, and even cardiac surgeries—when drawing conclusions [6, 8]. From a clinical perspective, this approach appears to be suboptimal.

Effective management of acute postoperative pain following thoracoscopic lung resection is critical, as it significantly influences recovery and is closely associated with the risk of developing chronic pain [50]. The results of this study demonstrate that in terms of pain management—specifically resting VAS, cough VAS at various postoperative time points, and postoperative morphine consumption—PVB significantly outperforms ESPB. This indicates that PVB offers superior comfort for patients undergoing thoracoscopic thoracic surgery, particularly for those requiring potent analgesia. PVB effectively mitigates discomfort and inadequate analgesia, thereby promoting earlier recovery and reducing hospital stay. This not only enhances patients' psychological well-being but may also accelerate postoperative recovery and contribute to

long-term health. However, despite its superior analgesic effect, the complexity of PVB may restrict its widespread use in resource-limited settings. Therefore, balancing analgesic efficacy with procedural complexity remains a critical consideration in clinical practice. Furthermore, the results of this study indicate that although the analgesic effect of ESPB is slightly inferior to that of PVB, its safety profile is superior. Therefore, ESPB may be considered a second-choice analgesic option following PVB. In clinical practice, ESPB offers a safer alternative, particularly for postoperative pain management following thoracoscopic thoracic surgery, especially in patients with coagulation abnormalities or those at high risk for nerve block complications. Additionally, it is a suitable option when the anesthesiologist's experience with nerve block techniques is limited.

This research systematically evaluated the analgesic efficacy of TEA, PVB, ESPB, SAPB, INB, and control groups based on data from 28 randomized controlled trials. To address the heterogeneity among studies, the research included an in-depth analysis of factors influencing postoperative pain scores, as well as comprehensive subgroup analyses and meta-regression, to ensure the robustness of the findings. Nevertheless, this study has certain limitations, with significant heterogeneity being a primary concern. This heterogeneity may arise from several factors. First, differences in perioperative care protocols could introduce bias. Second, variability in the dosage and concentration of local anesthetics used in nerve blocks likely contributes to heterogeneity, thereby limiting the interpretability of the network meta-analysis (NMA) results. Additionally, incomplete data during extraction may have impacted the accuracy of the findings. For example, some studies did not explicitly report means, standard deviations (SD), or 95% confidence intervals (CI), necessitating estimation through software or formulas. Moreover, certain studies lacked precise numerical data, requiring the use of Engauge Digitizer software to create simulated coordinate systems and extract the needed values. Future research should prioritize high-quality, large-sample randomized controlled trials to further validate these findings and determine the most effective local analgesic technique and optimal drug formulation.

CONCLUSION

TEA is considered less suitable for postoperative analgesia in patients undergoing thoracoscopic lung resection due to its significant side effects. Both PVB and ESPB serve as more appropriate alternatives. PVB provides superior analgesic efficacy compared to ESPB, and the analgesic effect of ESPB is insufficient to fully replace PVB. However, ESPB is associated with fewer side effects, making it a more favorable option in terms of safety. Therefore, in clinical practice, we recommend PVB as the first-line choice for postoperative analgesia in patients requiring strong pain relief; however, the procedure should be performed by experienced anesthesiologists. For patients with coagulation disorders or those at high risk for complications from nerve blocks, or when the anesthesiologist has limited experience with nerve block techniques, ESPB may be a safer and more appropriate alternative.

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Data Availability. The datasets generated during and analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest. Fang Yue, Yongye Xie, Xiangdong Chen, Ruifen Xu, Hui Wang, Ning Bai, Minna Hou, and Jiao Guo have nothing to disclose.

Ethical Approval. This article does not contain any new studies with human participants or animals performed by any of the authors.

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